Chapter 4

Annotation conventions for tone-related phenomena

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1 Introduction

As has already been mentioned in Maslinsky et al. (2026 [this volume]) (Section 4), the presentation of the tonal system of a language within the framework of the ThoT project consists of two parts: an analytical report based on the Questionnaire and one or more texts annotated for tonal phenomena. Therefore, each language chapter in Parts II–IV is supplemented by an annotated sample text. This chapter details the annotation system briefly introduced in Section 6 of Maslinsky et al. (2026 [this volume]).

The mark-up system detailed below has been primarily developed for the purposes of automated processing and computation of quantitative typological indices. However, it also allows authors to unambiguously capture and elucidate finer details of their analysis. For this reason, some contributors to this volume have chosen to also use elements of this system for illustrative examples within the body of respective chapters.

For the reader's convenience, a brief summary of the annotation marks used throughout the texts is given at the end of this Chapter (Section 7).

2 Lines of representation

The representation of a sample text minimally consists of five lines:

Dmitry Gerasimov, Kirill Maslinsky & Valentin Vydrin. 2026. Annotation conventions for tone-related phenomena. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 89–113. Berlin: Language Science Press. DOI: ??

1) The **citation line**. This line serves to make the text easily readable and recognizable for a human reader; it is devoid of any annotation marks specific to the ThoT project. When the texts are uploaded to the ThoT database, the citation line is not used for automated analysis, but still serves to display the text on the frontpage for the respective language. The authors/annotators are free to choose whatever format of representation they deem to be most familiar to experts on the language and/or general audience: orthography, transcription, etc. For texts taken from published sources, it makes sense to reproduce the representation of the source. At the authors' discretion, this line can be split into several; e.g., for a language with an established literary tradition, both orthography and transcription can be provided.

   In most cases, the citation line is segmented into morphemes with hyphens and other symbols provisioned by the Leipzig Glossing Rules (Comrie et al. 2008).

2) The **surface annotation line**, indicating tonal spans along with prosodic and (optionally) morphological boundaries.[1] This is discussed in more detail in Section 3.

3) The **underlying annotation line**, which captures the relations between morphemes and the tonemes in their lexical specifications. This is discussed in more detail in Section 4.

4) The **gloss line**, providing standard morpheme-by-morpheme glosses. Note that this line strictly follows the morphological segmentation in the **underlying annotation line**, not either of the first two lines (where it may in principle differ). This is discussed in more detail in Section 5.

5) The **free translation line**, providing translation of the sentence/period into English.

The first four lines are left-aligned vertically, word-by-word.[2] The free translation line follows separately after every sentence.

[1]The automated quantitative analysis will only take into account morphological segmentation in the underlying annotation line. However, indication of morpheme boundaries in either the citation line or the surface annotation line contributes significantly to the readability of the annotated text. We thus set up the requirement that morphological boundaries should be marked at least once in the first two lines.

[2]Note that 'word' here means the prosodic rather than the grammatical word, even though the applicability of the former notion to the level represented in the underlying annotation

The surface annotation line is strictly intended to reflect surface-level information (which tonal spans are aligned to which prosodic boundaries), while the underlying annotation line is concerned with underlying lexical specifications (which toneme is supplied by which morpheme). While derivational information, like specific tonal processes (spread, polarization, etc.), can sometimes be inferred from the correspondences between the two lines, it is never annotated specifically.

3 Surface annotation line

The following information is encoded in the surface annotation line:

- boundaries of prosodic units (syllables, prosodic words, etc.; Section 3.1, Section 3.5);
- (optionally) morphemic boundaries (obligatorily if not shown in the citation line; Section 3.2);
- boundaries of tonal spans (Section 3.3);
- grammatical tones (Section 3.4);
- stress (if relevant; Section 3.6);
- surface realizations of tones (Section 3.8).

Combinations of different annotation markings are specifically discussed in Section 3.7.

The surface annotation line is based on a surface-oriented transcription, where effects of all internal and external sandhi processes are taken into account (otherwise, correct syllabification would not be possible).

3.1 Prosodic boundaries

The following prosodic boundaries are marked:

- **Prosodic words** are separated by blank space.[3]

line is far from obvious. While the relationships between morphosyntactic and phonological wordhood may be rather complex, we assume that in any case morphemes can be grouped according to which prosodic word they correspond to at the surface level. We do not expect, for instance, that in any human language a prosodic word boundary would fall within a segmental exponent of a morpheme.

[3] If two morphosyntactic words (or more) constitute one prosodic word, their boundaries are shown as morphemic boundaries in both underlying and surface lines.

- **Syllables** are separated by dots, e.g. Soninke cáq.qá 'believe.GER'.

- For languages where the mora is relevant for tonal distribution, **morae** are separated by commas: Kugama hì,ŋ̀.ká.rī 'people', nā́,ā́.kī 'cow'. For this reason, it may be necessary to indicate vowel length with double letters in this line of representation.

- For languages where the prosodic foot is relevant for tonal distribution, **feet** are separated by vertical bars, e.g. Bambara fá.lá|-tɔ́ 'orphan'.

- For languages where **tonal phrases** are identified, they are enclosed in a pair of curly braces (see Section 3.5 below for examples).

Prosodic words and syllables are assumed to be universal and are marked for all languages. Other prosodic units may be absent or irrelevant for the tonal system and thus not annotated.

Mora, syllable, foot and prosodic word constitute a hierarchy. Whenever a higher boundary in the hierarchy is marked, all the lower-level boundaries are assumed and *should not be indicated*. See Section 3.7 for more details.

3.2 Morphological boundaries

Segmentation into morphemes can be provided. As mentioned in Section 2, morphological boundaries in the surface annotation line are only intended to improve readability and are not processed in the automated analysis.

Clitics can be contrasted to bound morphemes by using equals signs (=) instead of hyphens (per Rule 2 of the Leipzig Glossing Rules). Note, however, that some elements treated as clitics in language descriptions may exhibit properties of phonological words and are represented as such.

The use of angle brackets for infixes (per Rule 9 of the Leipzig Glossing Rules) is strictly avoided, as these are reserved for grammatical tones (Section 3.4) and will be processed accordingly. (This is one of the arguments in favor of providing morphological segmentation in the citation line rather than the surface annotation line.)

When a prosodic boundary and a morpheme boundary coincide, the prosodic boundary is marked first (see Section 3.7 for more details).

3.3 Tonal spans

3.3.1 The basics

Tonal spans are enclosed in square brackets, with the opening bracket immediately followed by the toneme label (typically a capital Latin letter or a combination of letters), e.g. Bambara [H sírá] 'road', Yue [LxH how^{25}] 'good'.

Here are the examples from Section 3.1 reproduced with the tonal spans indicated:

(1) a. Soninke [H cáq.qá] 'believe.GER'
 b. Kugama [L hì,ŋ̀.][H ká.]rī 'people', [H nʌ̃́,ʌ̃́.][M kī] 'cow'
 c. Bambara [H fá.lá|-tɔ́] 'orphan'

The tonal span of a toneme may coincide with a prosodic word (as in (1a–1c)), only involve a part of a word (not necessarily a morpheme) (1b, 2a, 2b, 2c), or span across word boundaries (2d). In the latter case, the opening and the closing bracket of the tonal span fall into different prosodic words.

(2) a. Hausa *màgàrúbà:* [L mà.gà.][H rú.][L bà,à] 'sunset'
 b. Lhasa Tibetan *ʦámʦám-là* [H ʦám.ʦám.-]là sometimes-in 'sometimes'
 c. Dom *kar-ǎ-l* ka.[LH r-à-l̄] see-FUT-1SG 'I will see'
 d. Wanga *βa-lá-kaluxan-a* *mu-ʃi-se*
 βà.-[H lá.-ká.lú.xá.n-á mú.-ʃí.-sé
 CL2.SU-NEAR.FUT-turn.around-FV CL18-CL7-time
 ʃí-tùutú
 ʃí.-][L tù,ù.][H tú]
 CL7-small
 'They will turn around in a moment.'

Parts of the segmental chain may fall outside any tonal span, like the suffix *-là* in (2b), the prefix *βà-* in (2d), or the initial syllable *ka* in (2c).

3.3.2 Floating tone realized as a downstep

There is a common phenomenon where a floating Low tone is realized as a downstep of the following High. For notational convenience, we mark the tonal span for the low toneme as zero-width (no segmental material inside) at the beginning of the following high toneme (3). To indicate that Low is realized as a downstep, an exclamation mark is put inside its tonal span.

(3) Bambara
jíri‘ *bɛ́ɛ*
[H jí.rí] **[L !]**[H bɛ́ɛ]
tree\ ART all
‘all the trees’

It is important to note that a downstep is not necessarily a manifestation of a toneme, and in some languages it does not constitute a tonal span. For instance, in Wanga downstep serves to demarcate the beginning of a tonal span of a High toneme when it follows another High in a phrase; therefore, it is not represented in the surface level annotation:

(4) Wanga
o-mú-límí *ꜜw-é*
o.[H mú.lí.mí] [H wé]
AUG-CL1-farmer CL1-3SG.POSS
‘his/her farmer’

3.3.3 Sharing of a syllable/mora between tonemes

The boundary between the tonal spans of two neighboring tonemes can fall inside a syllable or mora. In particular, one syllable within a longer tonal span can be shared with some other toneme. The recommended solution in this case is to keep the tonal spans separate and duplicate the segment that the tonemes share instead.

For example, in Bambara the specific article (a L tonal morpheme) can be realized on the last syllable of a high-toned word, producing a falling contour (5a). Since the MPU in Bambara is the syllable, and the relevance of the mora for the tonal system has not been independently demonstrated, moraic boundaries are not annotated and there is no option to mark up a monomoraic tonal span for L. Instead, the segment in question is duplicated without any prosodic boundary markers added, as in (5b):[4]

(5) Bambara
a. *jɛ́gɛ́* ‘fish’ + -$^{\text{L}}$ ‘ART’ ⟶ *jɛ́gɛ̂* ‘fish\art’
b. *jɛ́gɛ̂* [H jɛ́.gɛ́][L ɛ̀] ‘fish\art’

[4]For simplicity’s sake, (5b) omits grammatical tone marking, which is discussed in Section 3.4 below. The full annotation for this example is given there as (7a).

The fact that *ɛ* in the L tonal span does not constitute a separate syllable (or mora) follows from the absence of a separating dot (or comma), symbol of a prosodic boundary.

In other cases, where the tonal spans of both tonemes fit within a single syllable, one can say that the syllable is shared by the two tonemes. The tonal targets may be undershot in this situation, or even merge. We mark this kind of context with a single pair of brackets but with two tonemes listed in the beginning, separated with a semicolon:

(6) Babanki

Wùwì	*jì*^{L}	*ʃʉ̀*	*wàjn*^{H}	*à*	*mpfí.*
[L wùwì]	[L jì]	[L ʃʉ̀]	[L wàjn]	[H;L ā]	[H mpfí]
woman	P1	wash	child	for	mother

‘The woman washed the child for the mother’.

In (6), the floating H toneme at the right edge of *wàjn*^{H} ‘child’ docks rightward to the L-toned preposition *à* ‘to’, forming a falling contour which is simplified to mid. Two tonemes are realized on the monosyllabic preposition, which is represented as [H;L ā] in the surface annotation string. This will count as one syllable and two tonal spans and is equivalent to writing [H ā][L ā].

3.4 Grammatical tones

Grammatical tones are additionally marked with a pair of angle brackets enclosing their realization window. The latter may include a tonal span (7a, 7b, 7c, 8) or a sequence of tonal spans (7d); it may occupy a part of a word (7a–7b) or a whole prosodic word (7c–7d), or span across several words (8).

(7) a. Bambara *jɛ́gɛ̂* [H jɛ́.gɛ́]<[L ɛ̀]> ‘fish\art’

b. Jamsay *jùgɔ̂ː* [L jù.][H gɔ́,]<[L ɔ̀]> ‘know\ipfv’

c. Eastern Dan *ɗì.ʌ̏.ʌ̏* <[xL ɗì.ʌ̏.ʌ̏]> ‘mash\NEUT’

d. Soninke *ŋàlimaamí-ǹ* <[L ŋà.lì.màa.][H mí-n]> ‘imam\STCONSTR-DEF’

(8) Jamsay

ìjù jɛ̀m dùgú

<[L ì.jù	jɛ̀,m̀]>	[L dù.][H gú]
dog	black\HEAD	big

‘a big black dog’

As can be seen in (7–8), grammatical tone boundaries are always marked outside tonal span boundaries.

3.5 Tonal phrases

Tonal phrases (if identified in the language) are enclosed in a pair of curly braces (9–10).

(9) Dom

apalʌ	*gal(ʌ)*
{[LH à.pā,l̄]	ga,l}
woman	child

‘girl’

(10) Bunoge

nùmɛ́	*kòbálì*
{<[L nù.][H mɛ́]>	<[L kò.][H bá.][L lì]>}
hand\DEP	nail\HEAD

‘fingernail’

Just like grammatical tone boundaries (and unlike lower-level prosodic boundaries), tonal phrase boundaries are always marked outside tonal span boundaries. They are also placed outside grammatical tone boundaries.

3.6 Stress

Stress, if available in the language, is designated by an asterisk before the stressed syllable[5] (within the tonal span, but after all markers of other types of boundaries):

(11) Lithuanian

a. *pradúr-ti* pra.[H *dú,]r.-ti ‘pierce_through-INF’
b. *mir̃-ti* *mi,[H ŕ.-]ti ‘die-INF’
c. *výr-as* [H *ví,]i.r-as ‘man-NOM.SG’

(12) Ayutla Mixtec[6]

a. *šàtù=ì* [L *šà.tù.=][L ì] ‘trousers=1SG.POSS’
b. *šātų̀=į́* [M šā.][L tų̀.=][H *į́] ‘box=1SG.POSS’
c. *lášá=rà* [H lá.*šá.=][L rà] ‘orange=3SG.M.POSS’

[5]We follow Hyman (2006) in defining stress as always marking a syllable for the highest degree of metrical prominence, and do not recognize “moraic stress”. Thus, in (11c) the asterisk should be read as pertaining to the entire initial syllable (vi,i.), not just its first mora (vi,).

[6]Assuming that all three contrastive tone levels identified in (Pankratz & Pike 1967) are tonemic.

3.7 Relative ordering of annotation symbols

This subsection regulates cases when different annotation symbols have to be placed in the same position within the segmental string. This mostly happens when different types of boundaries coincide, which is a ubiquitous situation.

First, as mentioned at the end of Section 3.1, prosodic units constitute a nesting hierarchy. A prosodic word boundary always doubles as a syllabic boundary, a syllabic boundary implies a moraic boundary (if relevant), etc. Whenever a higher-level boundary is marked, lower-level boundaries are assumed (and automatically counted by the script) and *should not be marked*. Thus:

(13) a. Hausa *dó:lè* [H dó,o.][L lè] 'necessary', rather than *[H dó,o,.][L lè,.]

b. Bambara *bànfula* [L bàn|fù.là] 'cap', rather than *[L bàn.|fù.là.|]

Pleonastic symbols, as in the asterisked examples above, would lead to erroneous calculation of tonal statistics and so are avoided.

If morphemic boundaries are marked in the surface annotation string, it must be kept in mind that the morpheme does not form part of the prosodic hierarchy.[7] Thus, if a boundary of a mora, a syllable or a foot coincides with that of a morpheme, both are indicated, with the prosodic boundary placed first, e.g.: Bambara [H sɔ̀.rɔ̀|-là] 'receive-PFV.INTR'. However, we assume that any prosodic word boundary coincides with a morphemic boundary, so there is no need to put a hyphen at the end of a prosodic word (*[H sɔ̀.rɔ̀|-là-]).

For better readability, both morphemic and prosodic boundaries are marked inside tonal spans rather than between them. At tonal span edges, morphological and prosodic boundary markers are placed before the closing bracket rather than after the opening bracket:

(14) Musey
cām=bá
[L cā,-m̄.=**]**[H bá]   not *[L cā,-m̄][H .=bá]
wife-3SG.MASC.POSS=FEM
'his wife'

Overall, **the general ordering of annotation marks** placed in the same position within the segmental string is as summarized in (15) and illustrated in (16) (=12b):

(15) Prosodic boundary – Morphemic boundary – End of tonal span – Start of tonal span – Stress

[7] Morpheme and grammatical word form a hierarchy of their own. However, there are no provisions for marking grammatical word boundaries under the present annotation system.

(16) Ayutla Mixtec
šātṳ̀=į́
[M šā.][L tṳ̀.=]**][H** *į́]
box=1SG.POSS
'my box'

If grammatical tone (GT) boundaries are added to the mix, the full ordering is:

(15′) Prosodic boundary – Morphemic boundary – End of tonal span – End of GT – Start of GT – – Start of tonal span – Stress

Tonal phrase boundaries only occur at prosodic word boundaries (where morphemic and lower-level prosodic boundaries are never marked). As noted in Section 3.6, tonal phrase boundaries are always placed outside all other kinds of boundaries.

3.8 Surface realization of tones

To give the reader a better notion of the actual pronunciation of forms in the text, surface realizations of tones are notated in the surface annotation strings, using diacritics or Chao tone numbers (with 1 for the lowest level and 5 for the highest).

All tonal targets that serve as realization of tonemes within tonal spans are notated on every syllable or mora (for the languages where the MPU is the mora), as well as manifestations of default tone outside tonal spans. Intonational tones and surface pitches that result from pitch interpolation between adjacent tonal targets are not marked.

Allotonic variation is reflected if it is significant enough to be captured by the transcription used, i.e. involves variation in the composition of tonal targets. E.g. in Bambara *dàma-tɛ̀mɛ* [L dà.má.-][L tɛ̀.mɛ̀] 'exaggerate', where the first Low toneme is realized as LH and the second as L.

4 Underlying annotation line

The underlying annotation line primarily captures the relationship between tonemes and morphemes. This level corresponds to what is commonly understood as underlying tone.

4.1 M-items

The underlying phonological representation of a morpheme consists of two parts: a segmental component and a tonal component. In the underlying annotation line, morphemes are represented via notation units which we call *m-items*.

An **m-item** is an ordered pair of a tonal and a segmental specification of a morpheme, separated by a blank space and enclosed in parentheses. Thus, the notation in (17a) means that the morpheme (underlyingly) has the segmental composition *sira* and a High toneme. Both parts of a morpheme's underlying representation may be zero. Toneless morphemes have "0" written in place of tonal specification in their m-items (17b). Conversely, for tonal morphemes with no segmental specification, only the tonal part is indicated (17c).

(17) a. (H sira) ⟶ sírá
b. (0 ra)
c. (L)

Importantly, zero morphemes (with no overt expression whatsoever, whether segmental or suprasegmental) are not indicated in the annotation.

4.2 Contour tonemes and multiple tonemes in an m-item

When a morpheme has several tonemes in its underlying representation, **semicolons** are used to separate their labels in the m-item. Thus, in (18a) the Soninke morpheme bears two lexical tonemes, while the Dom morpheme in (18b) bears a single contour falling toneme, notated as HL.

(18) a. Soninke (H;L qulle) ⟶ *qúllè* 'white'
b. Dom (HL kai) ⟶ *kâi* 'to cry'

If a morpheme has several allomorphs that are not reducible to a single underlying form via general phonological rules of the language (including, but not limited to cases of suppletion), it is assumed to have several different underlying representations. For the treatment of morphemes whose exponents are operations rather than units, see Section 4.5.3.

4.3 Alignment point

The m-item notation is designed to be maximally agnostic about the specific ways in which tonemes associate to the segmental part of the specification of a morpheme. The m-item states only that the toneme(s) come from this particular morpheme, and that is all.

Yet in some languages there is a need for additional lexical information about the toneme's alignment in a morpheme to predict its surface realization. For instance, in Japanese, there is only a single falling toneme HL, but the position of the pitch drop in longer words is not predictable. We capture such information via the notion of **alignment point**, a lexical specification that a tonal target is coordinated with the edge of a prosodic constituent. The position of an alignment point is determined based on the position of the toneme in a **prototypical form** of the morpheme. The prototypical form is a surface form where the position of the toneme is not conditioned by other morphemes or intonation. This definition is not perfect and may need further elaboration; it is akin to a more traditional notion of lexical pre-association.

The alignment point is marked with a pair of ^ symbols: one in the tonal specification, and the other in the segmental specification, at the left edge of a tonal target. It indicates how tonal targets should be aligned relative to the segmental part.

For example, in Japanese, a language with only one toneme, /HL/, the alignment marks in the word *otama* 'ball' (19) can be read as follows. The right edge of the H target is aligned with the rhyme of the *ta* syllable, and the left edge of the L target is aligned with the onset of the *ma* syllable; we place the ^ symbol at the left edge of the L target (which coincides with the right edge of the H target).

(19) Japanese
(H^L ota^ma) ⟶ *òtámà*

Note that in this case the lexically important information is the position of the transition from high to low. The start and end of the realization of the entire falling toneme (the first mora of the word receives a default low tone and remains outside of the toneme span) are defined by the general rules of the language, and are therefore not marked in the m-item.

An alignment point is only indicated if the position of tonal target(s) is not predictable based on the general rules of the language (e.g., if the number of available syllables or moras exceeds the number of tonemes), as in (20a, 20b).

(20) Bambara

a. *búntèni*
[H bún][L tènì]
(H;^L bun^teni)
'yellow scorpion'

b. *nànselù*
[L nànsé][L lù]
(L;^L nanse^lu)
'grass *Cleome gynandra*'

When no alignement point markers are present within an m-item, this means that no contrast is available and/or facts about alignment can be established by general rules of the language. For example, when in Bambara a disyllabic morpheme is specified by two tonemes, /H/ and /L/, there is no need to indicate an alignment point (21), because this language has no disyllabic morphemes where the alignment point of the second (low) toneme would not coincide with the onset of the second syllable.

Bambara

(21) *sálòn*
[H sá][L lòn]
(H;L salon)
'last year'

On the other hand, indication of the alignment point is necessary for disyllabic bitonemic (L;L) morphemes in Bambara, because two models are available here, e.g. *mùmɛ̀* (L;^L mumɛ^) [L mù.mɛ́][L ɛ̀] 'entire', but *gwèlù* (L;^L gwe^lu) [L gwě.][L lù] 'African scops owl (*Otus scops*)'.

In principle, multiple pairs of alignment points can be used in the same m-item. The only condition is that the number of alignment points in the tonal and segmental parts of the specification **must be equal**.

4.4 Floating tones and phantom structures

Alignment points are also used to represent a situation where a tonal target is realized (always or in some contexts) outside of its m-item, i.e. on the neighboring segments. These phenomena are usually analyzed as floating tones or sometimes as tones defined on a phantom structure, see Rolle & Lionnet (2020).

For example, in Babanki a TAM marker *tə̀*^{L} is analyzed as having one inherent L toneme[8] and one floating /L/ on its right edge. For example, if this marker is

[8] By inherent toneme (in contrast to floating tonemes and grammatical tones) is meant a toneme which is associated with a morpheme and is realized on this morpheme when it appears in its prototypical form (i.e., when its realization is not conditioned by other morphemes and intonation).

followed by a disyllabic high-toned verb, the first syllable of the verb gets the low tone from the marker. This situation is notated in terms of m-items as follows:

(22) *tə̀* *sàŋlə́*
[L tə̀] [L sàŋ][H lə́]
(L;^L tə^) (H saŋlə)
P2 rejoice
'rejoiced'

The analysis in (22) presupposes that *tə* bears two distinct low tonemes: the first is inherent and the second one is lexically specified as floating.

There are cases when the alignment of a toneme relative to the segmental part of the morpheme is determined by counting the number of syllables/moras to skip. This kind of phantom structure can also be annotated using alignment points. For example, in Kuria, the remote future marker imposes a High toneme two moras to the right of the segmental part of the marker *re* up until the penultimate syllable of the word (23).[9]

Kuria (Rolle & Lionnet 2020: 2)

(23) *ntorehootóótéra*
n-tò.-rè.-hò,ò.<[H tó,ó.té.]> r-à
(0 n)-(0 to)-(^H re,,^)-(0 hootooter)-(0 a) ⟶
FOC-1PL-REM.FUT-assure-FV
'we will reassure'

Commas before the alignment point in the segmental part of the m-item indicate the distance of the realization window from the segmental co-exponent of the morpheme in moras. Use dots instead of commas to indicate distances in terms of syllables.

The marker of the noun class 10 in Babanki consists of a suffix *-sə́* with an inherent H toneme and another high toneme (a grammatical tone) realized at the beginning of the word stem (24).

(24) (L dʒom)-(H;^H ^sə) ⟶ *dʒóm*ꜜ*sə́* 'dream-CL10'

We may use the alignment point to indicate the stable position of the second high toneme on the suffix. But the position of the first high toneme depends on the length of the stem and thus it is not invariant relative to the segmental part of the m-item. No alignment point is marked for the first high toneme.

[9] In fact, the prefix *re-* with the H toneme realized on the stem is a "mixed grammatical morpheme", see Section 4.5.2.

4.5 Grammatical tones

4.5.1 Grammatical tonal morphemes

In the **underlying** markup, grammatical tonal morphemes are separated by means of a backslash from the segmental morphemes on which they are realized.

An example of a replacive tonal morpheme (for the Neutral Aspect) from Eastern Dan, in a four-line format (25):[10]

(25) gȍ
<[xL gȍ]>
(H go)\(xL)
sell\NEUT

An example of an additive tonal morpheme from Bambara:

(26) kólo‘
[H kóló]<[L ò]>
(H kolo)\(L)
bone\ART

4.5.2 Grammatical morphemes with tonal and segmental co-exponents

As mentioned in the Questionnaire (Section 7.1), a language can have morphemes with two co-exponents, segmental and tonal (grammatical tone); henceforth such morphemes will be referred to as “mixed morphemes”. There are two main types of mixed morphemes.

1) With a toneless segmental co-exponent, i.e., having no inherent toneme. For example, in Mwan the perfective affirmative marker has a toneless segmental component *-a/-ɗa*[11] and an autonomous L toneme co-exponent which is mapped on the verb stem. The grammatical tone may extend to the segmental co-exponent of its morpheme, in agreement with the general tonal rules of the language, as in (27), but it may also fail to be mapped on the segmental co-exponent (i.e. the suffix), as in (28), where the H toneme on *ɗáa* is generated through a polarization rule. Compare the underlying

[10] As mentioned in Section 2, we do not impose a standard for the representation in the first (“citation”) line; we leave it to the authors to make it as readable and transparent as possible.

[11] The choice of the segmental allomorph, *-a* or *-ɗa*, depends on the structure of the preceding foot, see Perekhvalskaya & Vydrin (2024).

representation of the perfective suffix ^{L}*-a/-ɗa* with a quasi-homonymous verbal derivative suffix *-ɗà* (glossed as MRPH) where /L/ is an inherent toneme which is replaced with /H/ through a polarization process in (28).

(27) gɔ̃̀-ã̀
<[L gɔ̃̀-.ã̀]>
(M gɔ̃)-(L^ ^a)
sell\PFV.AFF

(28) dù-ɗá-á
<[L dù|-]>[H ɗá-.á]
(M du)-(L ɗa)-(L^ ^ɗa)
stop-MRPH-PFV.AFF

2) A morpheme may have a segmental co-exponent provided with an inherent toneme, while additionally there is a separate tonal co-exponent, mapped, by default, outside the segmental co-exponent of this morpheme. In the underlying annotation line, capital letters designating both tonal components are separated with the alignment point marker, as in example (29), which reproduces example (24).

(29) Babanki
dʒóm꜌sə́
<[H dʒóm.-]>[L !][H sə́]
(L dʒom)-(H;^H ^sə)
dream-CL10

4.5.3 Morphological operations (process-based grammatical tones)

Both the segmental and the tonal component of a morpheme's exponent can be operations. To mark these up, we use conventional tags in the respective part of an m-item.

The most common segmental morphological operation is reduplication, which is tagged RDPL. Following the recommendation of the Leipzig Glossing Rules, the reduplication morpheme is separated with a tilde instead of a dash in both surface and underlying lines. (RDPL) would be an m-item for any morpheme expressed by reduplication with no additional tonemes introduced, as in (30), where the tonal melody of the original non-reduplicated form *gbɛ̋ɛ̏* 'difficult' extends to the entire reduplicated form.

(30) Eastern Dan
gbḗe~ gbȅe
[xH gbḗ.ḗ|~][xL gbȅ.ȅ]
(xH;xL gbee)~(RDPL)
difficult~ INT
'very difficult'

(H RDPL) may stand for reduplication deriving a deverbal noun in Yoruba, where a grammatical H toneme is imposed on the reduplicant (31).

(31) Yoruba (Downing 2011)
lí~là
[H lí.~][L là]
(H RDPL)~(L la)
GER~split
'splitting'

A special tag (SBTR) is reserved for subtractive tonal morphemes. E.g., in Japanese there is a mixed morpheme of adjectivization consisting of a segmental suffix *-teki* and a subtractive tonal co-exponent deleting the toneme on the stem (32).

(32) Japanese
keizai-teki
ke,i.za,i-.te.ki
(H^L ke^izai)-(SBTR teki)
economy-ADJ
'economical'

All other morphological operations are designated as OPER, a tag which may appear in either the segmental or the tonal part of an m-item. The m-item representing an operation is separated from the preceding m-item with a backslash (in both the underlying line and the gloss line). The actual tonal transformation performed by the morpheme operation is not described explicitly, but deduced from the correspondence between the surface and the underlying line. Some examples of various morphological tonal operations follow.

The Nominative marker in Kipsigis (Sande 2022: 416) is a tonological operation inverting all tonemes of the nominal base (L to H and H to L) (33).

(33) Kipsigis
mìntílì:l
<[L mìn.][H tí.][L lì:l]>
(H;L;H mintili:l)\(OPER)
sour.OBL\NOM

The JNT morpheme in Eastern Dan modifies the lexical tone of a verb in various ways, generally by lowering it (34).

(34) ɗō
<[M ɗō]>
(H ɗo)\(OPER)
go\JNT

The following example (35) shows a mixed Japanese morpheme (deriving names of shops and other establishments) consisting of a segmental suffix *-ya* and a tonal operation shifting the alignment point of the toneme /HL/ to the right edge of the stem.

(35) rjóòrì		rjòórí-yà
[HL rjó,ò.ri]	⟶	rjò,<[HL ó.rí-.yà]>
(H^L rjo^ori)		(H^L rjo^ori)-(OPER^ ^ya)
cuisine		cuisine-SHOP

4.6 Lexically specified stress

In languages with stress, its position can also be lexically specified. The underlying position of stress is marked inside the segmental part of the respective m-item with an asterisk placed before the accent-bearing syllable:[12]

(36) Salawati Ma'ya (Remijsen 2002: 48)

a. *ˈmana*^{3}
*mā̃.[H ná]
(H *mana)
'light (of weight)'

b. *maˈna*^{3}
mà.[H *ná]
(H ma*na)
'grease'

[12]Of course, many languages may be argued to have no syllables in the underlying representation, but then one would not expect lexically specified stress in such languages.

Note that this mark is not used if the position of stress is predictable from tonal alignment. For instance, in Lithuanian (ex. (12) above), a syllable is stressed by virtue of containing a H-toned mora, so there is no need to indicate its position within the m-item.

4.7 Conjunction of m-items

To match the surface annotation line, the underlying annotation line is conventionally segmented into prosodic words. M-items corresponding to the same prosodic word are joined by hyphens and other symbols of morphological boundaries envisioned in the Leipzig Glossing Rules. These are placed outside the parentheses enclosing m-items.

4.7.1 Morphological boundaries

By default, m-items are joined by hyphens, as in (27) above. Other morphological boundary markers used in the underlying annotation line (and, by extension, the gloss line) are as follows:

- Clitics can be contrasted to bound morphemes by using equals signs (=) instead of hyphens (per Rule 2 of the Leipzig Glossing Rules).
- The tilde (~) is used as a boundary marker between the reduplicant and the base (per Rule 10 of the Leipzig Glossing Rules), see (30, 31) in Section 4.5.3.
- Tonal morphemes that have no segmental part in their exponent follow the morpheme on whose segments they are realized, joined by a backslash (\) (adopted from Rule 4D of the Leipzig Glossing Rules). This concerns both unit-based tonal morphemes (37) and tonal morphological operations (38):

(37) Cantonese
jyn^{55}
[xH jyn^{55}]
(LH jyn)**(xH)**
far\\INTNS
'very far'

(38) Eastern Dan
kʌ̀
<[xL kʌ̀]>
(M kʌ)\\(OPER)
do\\JNT

Tonal morphemes can be realized on segments belonging to several different morphemes (or even words). In such cases it is up to the author/annotator where to place the respective m-item. Generally, it is written after the last affected morpheme, as in (39) and (40), but depending on the language, other ways to linearize the m-item may be preferable, cf. (41).

(39) Jamsay

ñɛ̀ː-w^{n}ɛ̀

<[L ñɛ̀,ɛ̀.-w̃ɛ̀]>

(H ñɛː)-(H wV)**(L)**

eat-CAUS\\PFV

'he made him eat'

(40) Jamsay

ìjù	*jɛ̀m*	*dùgú*
<[L ì.jù	jɛ̀,m̀]>	[L dù.][H gú]
(LH iju)	(H jɛm)**(L)**	(LH dugu)
dog	black\\HEAD	big

'a big black dog'

(41) Iquito

núú	*nùù=k^{w}aata-ki*
<[HL nú,ú	nù,ù.=]> k^{w}a,a.ta.-ki
(0 nuu)**(HL)**	(0 nuu)=(0 k^{w}aata)-(0 ki)
3.GNL.PRO\\IRR	3.GNL=clear.weeds-NPST

's/he will clear it of weeds'

In (39) from Jamsay, the tonal Perfective morpheme is realized on a verbal stem consisting of two morphemes. In (40), also from Jamsay, the realization window of the grammatical tone includes a noun phrase consisting of two words; the tonal morpheme is only glossed once, after the second word.[13] In (41) from Iquito, the tonal Irrealis morpheme also affects two different words, but its realization window does not cover them fully. It can be viewed as a segmentally zero lexeme occupying the position between the two words; but since it does not form a prosodic word of its own, here it is conventionally joined to the first word, the pronoun.

[13] Alternatively, the first two words could be glossed as "dog\\HEAD black\\HEAD", in accordance with the common practice. However, we consider the replacement of the lexical tones of these words as a realization of a single grammatical tonal morpheme (and a single tonal span), and marking HEAD twice would result in an incorrect automatic morpheme count.

4.7.2 Discontinuous morphemes

M-items in the underlying annotation line are presented sequentially and cannot be nested. Infixes and transfixes are written after the morphemes that they interrupt (or before, if they are clearly left peripheral) and joined with a hyphen (angle brackets, as per Rule 9 of the Leipzig Glossing Rules, are not used).

Example (42) illustrates the verbal infix -$^{\text{L}}$*g*- participating in the formation of some tense-aspect forms in Eton, while (43) shows the degree modifying transfix -*k*-…-*k*- in Eastern Dan.

(42) Eton (van de Velde 2008: 368)
mè-sîl<g>à
[L mè.-][H;L sîl.-][L g-à]
(L mə)-(H;L sila)-**(L^ ^g)**
1SG-ask-G

(43) Eastern Dan
ɓlű<k>űɓlȕ<k>ȕ
[xH ɓlű|-][xH k-ű|][xL ɓlȕ|-][xL k-ȕ]
(xH;^xL ɓluu^ɓluu)-(0 k-k)
pointless-INTENS

Circumfixes are treated as combinations of two separate morphemes, each represented by its own m-item. At present we are unaware of any instances of circumfixation in tonal languages for which a bimorphemic analysis would not be possible (and even preferable).

5 Gloss line

This line consists of morpheme-by-morpheme glosses. Crucially, the number and sequence of glosses in this line (and of separators between them) mirror exactly the number and sequence of m-items in the underlying annotation line (and of separators between them).

This means, inter alia:

- use of hyphens, equals signs, tildes, and backslashes (but not angle brackets) as separators, per Section 4.7.1;
- linearization of glosses for tonal morphemes with multi-word realizations, per Section 4.7.1;

- linearization of glosses for infixes and transfixes, per Section 4.7.2.

Cf. examples (39–41) in the preceding section, where the glosses reflect the morphological segmentation in the underlying annotation line, but not necessarily that found in the citation line.

Sometimes a lexical item may prosodically form two or more separate words, some or all of which never occur on their own. For instance, in Iu Mien, *Yih Bunc* 'Japan' is a borrowed proper name, in which neither of the two parts can occur outside of this combination and be assigned its own meaning. In such cases, each prosodic unit is annotated as a separate m-item, with the translation of the whole simply repeated for each part in the gloss line:

(44) Iu Mien

Yih	*bunc*
[ML i]	[L pun]
(ML i)	(L pun)
Japan	Japan

'Japan'

6 Full sentential examples

Here are two full sentential examples with a complete markup:

(45) Kugama

ɗì	ɔ́ɔ=si=ɛ=^{L}	puy	ɓáa=^{L}	bòm
[L ɗì]	[H ɔ́,ɔ́.=]<[L s=ὲ]>	pū,ȳ	[H ɓá,á]	<[L !]>[L bò],m̄
(L ɗi)	(H ɔɔ)=(0 si)=(0 ɛ)\(L)	(0 puy)	(H ɓaa)\(L)	(L bom)
3PL.HUM.FUT	dig=APPL=NMLZ\ GEN	hole	in\ GEN	hut

'They will dig a hole inside the hut.'

(46) Dom

^{LH}*none*	^{H}*nul*	^{HL}*du*	^{H}*ku*	^{HL}*el-e*
[LH nò.né]	[H nú,ĺ]	[HL dú,ù]	[H kú]	[HL é.l-è]
(LH none)	(H nul)	(HL du)	(H ku)	(HL el)-(0 re)
1NSG.INC	river	frog	catch	(make)-conj(ss)

^{HL}*bus*	^{LH}*kuna*	^{H}*i*	^{H}*ka*	^{LH}*kopa*	^{H}*s-re*
[HL bú,s̀]	[LH kù.nā]	[H í]	[H ká]	[LH kò.pā]	[H s-ŕ.é]
(HL bus)	(LH kuna)	(H i)	(H ka)	(LH kopa)	(H s)-(0 re)
bush	around	dem	bird	bird	hit-conj(ss)

el-^{LH}*a-pga*	^{H}*er*	^{HL}*si*	^{HL}*su-gwa*
e.<[LH l-à,-p̀.gá]>	[H é,ŕ]	[HL sí,ì]	[HL sú.-gwá,l̀]
(HL el)-(LH a)-(LH pn)-(0 ka)	(H er)	(HL si)	(HL s)-(0 m)-(0 kal)
make-fut-1pl-srd	to	be.bush	(hit)-3sg-loc

^{LH}*na-pn=*	(^{H})*wa*	^{HL}*du-m*	^{HL}*du-gwe*
{<[LH nà.-pń]>	wá}	[HL dú,-m̀]	[HL dú.-gwè]
(HL p)-(LH ra)-(LH pn)	(H wa)	(HL d)-(0 m)	(HL d)-(0 m)-(0 ke)
go-fut-1pl	enc.wa	say-3sg	say-3sg-ind

‘Let’s go to the bush to catch frogs by the river and to catch birds in the woods.’ (adapted from (Tida 2006: 255–256))

7 Summary of annotation marks

In the **surface annotation line**:

[T ...]	tonal span (of the toneme *T*);
[T1;T2 ...]	a MPU shared by two tonemes;
[L !]	a zero-length tonal span of a L toneme realized as a downstep;
.	syllabic boundary
,	moraic boundary
|	prosodic foot boundary
<...>	grammatical tone
{...}	tonal phrase
*	stress (indicated before the stressed syllable)

In the **underlying annotation line**:

(T s)	m-item, a lexical specification of a morpheme with a segmental part *s*, and a tonal part *T*;
;	separator between labels of tonemes belonging to the lexical specification of the same morpheme;
^	alignment point;
rdpl	reduplication;
SBTR	subtractive tonal operation;
OPER	any other tonal operation;
\	separator of a grammatical tonal morpheme;
~	separator between the reduplicant and the base.

8 Abbreviations in glosses

ADJ	adjectivizer
APPL	applicative
ART	article
AUG	augment
CAUS	causative
CL1…CL18	nominal/ agreement class marker
CONJ	conjunctive
DEF	definite
DEM	demonstrative
DEP	syntactic dependency marker
DS	different subject
ENC	enclitic
FEM	feminine
FUT	future
FV	final vowel
G	*g*-suffix/infix (in Eton)
GEN	genitive
GER	gerund
GNL	general number
HEAD	syntactic head marker
HUM	human
INC	inclusive
IND	indicative
INF	infinitive
INTNS	intensifier
IPFV	imperfective
IRR	irrealis
MASC	masculine
NEAR.FUT	near future
NEUT	neutral aspect
NMLZ	nominalizer
NOM	nominative
NPST	non-past
NSG	non-singular
OBL	oblique
PFV	perfective
PL	plural
POSS	possessive
PRO	pronominal
SG	singular
SHOP	derivational suffix for names of shops
SRD	subordinate
SS	same subject
STCONSTR	status constructus
SU	subject

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